Curbing profligacy in materiel acquisitions a challenge

RAHUL BEDI
SENIOR JOURNALIST

THE multiplicity of platforms employed by India's military poses a massive, hugely expensive and complex logistical challenge, one that increasingly consumes the armed forces' annual revenue budget, which has consequently grown steadily in recent years.

Unlike other analogous militaries that have greater homogeneity and commonality in their assets, India's three services remain profligate in their choice of haphazardly acquired assorted weaponry, with scant effort at standardising equipment to effect savings on sustainment. Instead, all their diverse gear not only trigger mounting revenue allocations each year but, worse, lead to platform cannibalisation in which components or sub-assembles are removed from an active, in-service resource and installed on another, rendering the former inoperative.

Furthermore, this burgeoning equipment array also mandated the training and sustaining of a vast number of technicians and mechanics to service and maintain it for its average operational lifespan of around 25-30 years. A cross-section of service personnel, veterans and defence analysts concurred that if the military streamlined its kit profile, its yearly capital budget allocation, which averaged around 40 per cent of the overall defence outlay, could see an upsurge, augmenting thereby force modernisation and enhancing operational efficiency.

From 2021-22 to 2023-24, the 'stores' allocation that pertains to equipment spares, components and replacements for the Army has risen from Rs 20,383 crore to Rs 23,965 crore, for the Navy from Rs 6,054 crore to Rs 9,221 crore and for the Indian Air Force (IAF) from Rs 9,679 crore to a whopping Rs 17,200 crore.

The IAF, for instance, operates seven fighter types, including Russian Sukhoi Su-30 MKIs, upgraded MiG-29M and MiG-21 Bisons, French Rafales and Mirage-2000Hs, Anglo-French SEPECAT Jaguars and the indigenous Tejas Light Combat Aircraft. It also plans to acquire 114 multi-role fighter aircraft (MRFA), which could add yet another combat-type aircraft to the IAF's existing portfolio.

Similarly, the IAF's six transports comprise Russian and Ukrainian Ilyushin Il-76s and Antonov An-32s, C-17s and C-130J-30s from the US, legacy Avro HS-748s from the UK and licence-built Dornier 228s from Germany. Moreover, the induction of 56 Airbus Defence and Space C-295MW transporters, of which 16 are being acquired in flyaway condition and the remaining 40 licence-built locally, was imminent, making it the IAF's seventh such aircraft.

EQUIPMENT PROFILE: The Navy deploys three dissimilar conventional diesel-electric submarines. FILE PHOTO

The forces' helicopter fleet incorporates seven platforms including AH-64 Apaches and CH-47 Chinooks (US), Mil Mi-17 variants and Mi-24s (Russia), indigenous Dhruv advanced light helicopter variants and licence-made French-origin Aerospatiale Alouette IIIs (Chetaks) and Aerospatiale SA-315B (Cheetahs).

The Army's jumbled equipment buys are set to render it possibly the world's only army which could, over the next decade, field five main battle tank (MBT) types simultaneously. At present, it operates Russian T72M1 'Ajeyas and T90S 'Bhishmas, alongside indigenous Arjun Mk1/1As, but is also on track to develop Zorawar, a 25-tonne tank, in addition to procuring 1,770 Future Ready Combat Vehicles by 2030.

> The allocation pertaining to equipment spares, components and replacements has been on the rise.

The US, in comparison, operates M1 Abrams variant MBTs, while Israel fields the Merkava, Britain the Challenger 2, Canada the Leopard 2A6M, Germany the Leopard and France the Leclercs. Conversely, the Russian army has T72, T80 and T90 MBTs — all with a high degree of commonality — and has only recently begun inducting the T-14 Armata, predicated on a multi-faceted Universal Combat Platform concept.

Additionally, the Army's seven artillery guns include M777s (UK/US), FH-77Bs (Sweden), local Dhanush howitzers, Soviet-era 130mm M46's and the indigenously upgraded Sharang M-46s, domestically sourced 105mm field guns and more recently the K-9 Vajra self-propelled howitzers, built jointly in India by Larsen & Toubro and South Korea's Hanwha Defence. The Army is also in the process of acquiring the domestically designed and manufactured 155mm/52 calibre Advanced Towed Artillery Gun System, besides evaluating the Israeli Autonomous Towed Howitzers Ordnance System for procurement.

The Army also deploys at least six helicopter types such as Apaches and Dhruv variants, including Prachand and Rudra attack models, indigenously designed light utility helicopters, Cheetahs and Chetaks. Its assault rifle profile encompasses TAR-21s and Galils from Israel, SIG716s from the US, AK-47s and AK-203s from Russia alongside the local Indian Small Arms System, in addition to six types of sniper rifles from Russia, Israel, Germany, Italy/Finland and the US. Four anti-material rifle types, at least five different kinds of machine guns/pistols from Belgium, Germany, Israel, Switzerland and the US and three models of air defence guns from Soviet Union/Russia and Sweden, among other varied equipment, all of which need servicing and back-up.

The Navy, relatively frugal among India's three services in its equipment profile, deploys two diverse aircraft carriers and three dissimilar conventional diesel-electric submarines or SSKs: Russian Type 877 EKM Kilo-class boats, German HDW Type 209/1500 Shishumar-class SSKs and French Kalvari-class (Scorpene) platforms. Remarkably, the Navy is evaluating yet another SSK — to be sourced locally under transfer of technology to a local shortlisted shipyard from Germany or Spain — which is scheduled to join service in the early 2030s, making it the fourth submarine model in the forces' growing platform stock list.

The fundamental flaw lies in defence planning, despite the embarrassment of riches of high-powered oversight decision-making bodies created to sustain India's higher strategic and military realms. The remit of the Defence Planning Committee (DPC), set up in 2018 and headed by National Security Adviser Ajit Doval, is the 'overarching' body to manage the country's security strategy, prepare military capability plans and fast-track and rationalise materiel acquisitions.

DPC members include the Chief of Defence Staff, the three service chiefs and the defence and foreign secretaries. But so far, they all collectively appear unperturbed — or perhaps unwilling — to trigger platform and equipment streamlining and reorganisation, and are simply content to maintain the status quo.

Why Rafale could be IAF's default MRFA option

RAHUL BEDI
SENIOR JOURNALIST

A HOST of recent developments have placed Dassault Aviation, France's principal combat aircraft manufacturer, in pole position to potentially secure the IAF's long-pending requirement for 114 multi-role fighter aircraft (MRFA) with its twin-engine Rafales.

Joining the dots, a cross-section of military veterans, defence analysts and industry officials have posited that backed by growing Rafale sales to India and Dassault's inclination to transfer fighter and related technology to it, the IAF could well end up with Rafale as its default MRFA option.

If so, it would be an ironical redux of the Ministry of Defence's stillborn 2007-08 tender for 126 Rafales — of which 108 were to be licence-built by Hindustan Aeronautics Limited — that was scrapped in 2015 due to contractual, political and bureaucratic snafus.

A review of these emerging dealings with regard to the possible induction of additional Rafales into the IAF is instructive on multiple counts. At the outset, Dassault will supply 26 Rafale-M (Marine) fighters to the Indian Navy (IN) for deployment aboard INS Vikrant, its newly commissioned aircraft carrier. The multi-role carrier-borne fighter's (MRCBF) 'commonality' with the 36 Rafales, which the IAF had imported in 2016 for $9 billion, had influenced the IN's choice following user trials in 2022, which featured the rival Boeing F/A-18 Block III 'Super Hornet' fighter.

The delivery of these 26 fighters over the next two-three years would total an inventory of 62 Rafale variants in India's arsenal, not an insignificant number. Hence, increasing this quantity further, according to some retired senior fighter pilots, to meet the IAF's requirement for 114 MRFA makes 'immense operational, commercial and logistical sense.'

Dassault, they said, had already established a Rafale flight training and Maintenance, Repair and Overhaul (MRO) facility at Ambala Air Force Station, which would go a long way towards reducing the overall costs for any additional buys. Importantly, acquiring tried-and-tested Rafales would hasten fighter inductions by dispensing with trials and boost the IAF's declining fighter squadrons, whose number has depreciated to around 29 from a sanctioned strength of 42.

They added that acquiring extra Rafales would also streamline the IAF's diverse fighter catalogue, which currently features seven aircraft types, sustaining all of which is an enduring logistical nightmare and a costly affair for the force.

BOOST: Dassault will supply 26 Rafale-M fighters to the Indian Navy for deployment aboard INS Vikrant. ANI

In a related development that could enhance Dassault's MRFA bid, the French manufacturer is believed to be in advanced negotiations to acquire 51 per cent stake of its partner Anil Ambani in Dassault Reliance Aerospace Limited (DRAL) joint venture in Nagpur. India permits 100 per cent foreign direct investment in individual cases and Dassault is reportedly keen on acquiring DRAL, which, if effected, would augment its chances of bagging the MRFA deal.

At present, Dassault owns 49% of DRAL, formed within days of India confirming the IAF's 36 Rafale purchase, to discharge the 50% offset obligation of the inclusive contractual price in accordance with MoD's procurement procedures.

> Acquiring tried-and-tested Rafales would speedily boost the strength of IAF's fighter squadrons.

Initially, DRAL was tasked with producing components for Dassault's Falcon business jets and only recently, it had begun producing sub-assemblies like engine doors and canopies for Rafales. But as per media reports, a domestic financial resource crunch had curtailed DRAL's manufacturing capabilities, rendering it vulnerable to a buyout.

Meanwhile, the MRFA procurement envisages the importing of a squadron of 18 shortlisted fighters in flyaway condition from amongst seven models proposed by overseas original equipment manufacturers (OEMs) in response to the IAF's April 2018 Request for Information (RFI). The remaining 96 platforms would be built indigenously by a collaborative venture between the qualified OEM and a domestic strategic partner (SP) from either the private or public sector, with progressively enhanced levels of indigenisation in the all-encompassing deal valued around $25 billion.

The MRFA tender is expected imminently, industry sources said, with the selected platform required to complete 30-35 years of squadron service or flying time of 6,000 hours, with at least one midlife upgrade. Senior IAF officers estimated that MRFA numbers could increase to around 200 units for the IAF alone, in addition to possible export options, resulting in the platforms' cost amortisation.

The six other OEMs which had responded to the IAF's MRFA RFI are Eurofighter Typhoon, Sweden's Saab (Gripen-E), Russia's United Aircraft Corporation and Sukhoi Corporation (MiG-35 'Fulcrum-F' and Su-35 'Flanker-E') and the US' Boeing and Lockheed Martin (F/A-18 and upgraded F-21).

However, in the light of the ongoing war in Ukraine, evaluating the two Russian fighter types for eventual IAF acquisition was, justifiably, irrational, considering the grave spares and components crisis the force is facing with regard to its fleet of 260 multi-role Sukhoi-30 MKIs and 50-odd upgraded MiG-29M fighter-bombers.

Alternatively, shortlisting the Typhoon would only mean adding to the IAF's continuing logistic challenges, whilst the US' F-18 and F-21 — a retrofitted F-16 — had been rejected by it on multiple capability counts during trials conducted 2010 onwards for the binned MMRCA contract. Saab's Gripen-E, on the other hand, was a single-engine platform, and though the MRFA RFI had not specified any preference for single or dual power packs, the IAF's intrinsic preference for the latter remains unstated.

Hence, by the process of elimination, Rafale was more than favourably placed in the MRFA sweepstakes, due not only to its operational superiority over its competitors, as acknowledged by the IAF — and now by the IN — but also the host of multiple ancillary factors.

There was also the abandoned contractual template for the MMRCA contract, which, industry officials said, could easily be 'tweaked' to suit an analogous MRFA purchase by resolving earlier glitches and shortening negotiations. These anomalies had centred on insistence by the MoD that Dassault shoulder eventual quality control responsibility for the 108 Rafales licence-built by HAL. This unwarranted conditionality had emerged as the deal-breaker for the MMRCA deal and led to the IAF procuring just 36 Rafales in flyaway condition, all of which were delivered by late 2022.

Even geopolitically, Indian diplomats and security officials conceded that it was less 'arduous' conducting materiel commerce with Paris than with Washington as the former was more flexible and pragmatic than the latter, especially with regard to transferring hi-tech military knowhow.

Non-lapsable Defence Modernisation Fund no silver bullet

AMIT COWSHISH
EX-FINANCIAL ADVISER, ACQUISITION, MoD

RAHUL BEDI
SENIOR JOURNALIST

THE issue of instituting a rollover or non-lapsable Defence Modernisation Fund (DMF) to upgrade India's three Services, mooted in 2004, remains a work in progress 19 years later. Replying to a question in the Lok Sabha last month, Minister of State (Defence) Ajay Bhatt stated that a 'special mechanism' was underway to operationalise the DMF, but did not elaborate when this long-promised undertaking would be put in place.

The decision to introduce a Rs 25,000-crore non-lapsable DMF was first announced by then Finance Minister Jaswant Singh in the NDA government in his interim 2004-05 Budget, to ensure 'committed availability' of monies for materiel procurements. At the time, Jaswant Singh had argued that a DMF would 'commit availability of adequate funds' as the process of defence procurement often extended to over three years. The fund was to be operationalised from the ensuing financial year.

But the NDA lost the General Election soon after, and the DMF provision was summarily dropped in the Budget presented by Jaswant Singh's successor from the Congress-led UPA that assumed office in mid-2004.

This should have sounded the death knell for the DMF, but it captured the imagination of successive Parliamentary Standing Committees on Defence (SCoDs), which continued to doggedly pursue it and demand its imminent establishment. The SCoDs insisted on a DMF with 'committed allocations' for a five-year period since 'lapsing of funds year-after-year (had) greatly hampered (the) procurement of defence equipment and modernisation of the defence forces'.

The proposal for creating a non-lapsable DMF had emerged following the Services' inability to annually expend their capital budget allocated for modernisation which, between 1999 and 2004, had cumulatively amounted to Rs 20,217 crore, or nearly 22 per cent of the overall outlay committed to this end. This lapse, however, was due primarily to the military's ineptness in fast-tracking acquisitions, of which their vast 'overreach' in equipment specifications was one of the several culpable shortcomings.

But badgered over years by an influential number of serving and veteran military personnel, the SCoDs erroneously and somewhat naively assumed that this 'surrender' of unspent balances at the end of each fiscal could be circumvented by transferring these amounts to a non-lapsable DMF. This prospect also excited the armed forces, who too mistakenly believed that such a fund could be dipped into at will, much like an ATM, to effect acquisitions over and above the allocated annual defence outlay.

FLAWED: Mooted in 2004, the DMF is not the panacea for the problems concerning military modernisation. ANI

This belief, however, remains flawed for multiple systemic reasons.

Constitutionally, all revenue from taxes, loans, disinvestment and analogous sources, aggregated by the Government of India, accrues to the Consolidated Fund of India (CFI), and no monies can be appropriated from it, even for beefing up non-lapsable funds, without parliamentary approval. This is how other non-lapsable funds, like the one for the northeastern states, work and the reality is that Parliament cannot be bypassed to replenish, access or establish such funds at will.

> Implementing the DMF would be tantamount to duplicating prevalent budgeting procedure, but expecting a different outcome.

These contrary arguments were presented by the Defence and Finance Secretaries to the SCoD between 2004 and 2006, but to no avail, and the latter continued to insist on founding the DMF. In fact, in 2017, the SCoD reiterated that it was 'imperative' to institute such a fund 'for enhancement and heightened operational preparedness' of India's military.

Thereafter, PM Narendra Modi's BJP-led government responded by referring the matter to the Fifteenth Finance Commission (FC), which too recommended the DMF's formation in its 2020 report, but its implementation obviously stands deferred. This is so probably because the DMF's overall rationale and utility were still being internally debated, even though there was no legal impediment in its founding.

The FC had also recommended that though the bulk of the DMF's corpus would be sourced from the CFI, it would marginally be supplemented by disinvestment in Defence Public Sector Undertakings (DPSUs) and via monetisation of surplus defence land, with an annual cap of Rs 51,000 crore on all such transfers. But analysts pointed out that since DPSU disinvestment and land monetisation could not be a regular revenue source, the burden of underwriting the proposed DMF would then inevitably fall to the CFI which, ironically, was the original source for annual budgetary allocations anyway, bringing matters back to square one.

In the meantime, the original rationale for launching the DMF had become infructuous, as the Ministry of Defence (MoD) had largely been expending its annual capital outlay, as evident from the available data for three successive years between 2019 and 2022. This, in turn, obviated the requirement for introducing a rollover account.

Alongside, elementary monetary logic too dictated that if the CFI, in the first instance, was capable of financing the non-lapsable fund, surely it could afford to make budgetary allocations of comparable amounts each year without routing it through a DMF, considerably complicating matters by adding yet another layer of bureaucracy to an already hidebound structure.

While deposing before the SCoD in 2006, then Finance Secretary Adarsh Kishore had summed up the DMF conundrum by stating that though no major conceptual problem existed in setting it up, the armed forces would not be 'particularly happy' if it was initiated, as all existing procedures that related to the yearly budget would, perforce, apply to operate it since Parliament could not be bypassed in this regard. Therefore, implementing a DMF would be tantamount to duplicating the prevalent budgeting procedure, but miraculously expecting a different outcome in its add-on DMF *avatar*.

Hence, it's little wonder that the DMF proposal remains stillborn, as few in the government have any inkling regarding its final contours. But it's immutable that DMF in any form would have limited, if any, use as the bulk of its finances would inevitably come from the CFI and their utilisation would always be subject not to MoD's discretion, but to parliamentary sanction.

In short, the DMF is not the panacea for the problems concerning India's military modernisation, but something more fundamental is — addressing the enduring financial resource crunch.

Decolonise armed forces, but with sensitivity

RAHUL BEDI
SENIOR JOURNALIST

THE 'decolonisation' of India's military in the ongoing 'Amrit Kaal', as decreed by the BJP-led Central Government, continues apace. At the recently concluded biannual Indian Navy (IN) commanders' conference in New Delhi, there were deliberations on sundry operational matters and on incorporating the hitherto proscribed *kurta-pyjama* as an accepted dress form in naval messes, wardrooms, official establishments and on formal occasions.

Astonishingly, the IN displayed possible variations of its proposed dress at the three-day conference for approval by Minister of State for Defence Ajay Bhatt; it included a mannequin exhibiting a shorter *kurti*, worn under a waistcoat, and a drainpipe *pyjama*. A cross-section of naval veterans concerned concurred that it was merely a matter of time before *kurta-pyjama*, in some configuration, joined the IN's formal couture alongside lounge and Jodhpuri suits and the striking 'Red Sea' rig, comprising black trousers, half-sleeve white shirt with rank-stripe epaulettes, black shoes and socks and cummerbunds of the same colour.

In recent months, the IN has taken the lead over the other two services in shedding its colonial mantle, in keeping with the government's goal of securing *atmanirbharta* or indigenisation in the armed forces, not only with regard to their materiel, but also their dress codes, rituals and traditions.

The Indian Army was established by the East India Company in the 18th and 19th centuries, while the Royal Indian Navy (RIN) and the Royal Indian Air Force (RIAF) came into being in 1934 and 1932, respectively, becoming the IN and the IAF after Independence. Understandably, all three embraced and, over decades, perpetuated many of their progenitors' hoary customs, conventions and practices which, in most instances, constitute the bedrock of most of the world's militaries.

But such indulgences are apparently anathema to the government. Consequently, in July the IN discontinued the Royal Navy (RN) practice of its senior officers carrying batons on the grounds, saying that it did not suit the 'transformed Navy of Amrit Kaal'. This was done shortly after its ensign or flag was indigenised and shorn of its colonial antecedents that had featured the blood-red Cross of St George for decades. Concurrently, the Navy also got a new *atmanirbhar* President's Standard and Colour and a revised *desi* crest.

EVOLUTION: The Indian Navy has taken the lead over the other two services in shedding its colonial mantle. PTI

Grizzled navalists conceded that the IN, of all the country's three services, had, because of its founding antecedents, many commonalities with the RN, not only in operational, logistic and doctrinal procedures, but also in its maritime mores, outlook and even superstitions. Over decades, the Navy's overriding modus vivendi and ambience continued to overtly display RN influences handed down by succeeding generations of officers. In fact, many RN officers visiting IN ships and establishments invariably found the atmosphere 'quaintly' English, right down to the drinks, cocktails and fare served in wardrooms, which often included roasted mutton, *papadams* and sweet mango chutney.

> Such arbitrary moves engender a backlash, especially among veterans, who are the backbone of the services.

Alongside, the Indian Army's Adjutant General branch, too, has embarked on ending 'archaic and antiquated' colonial traditions, dress codes, pipe and drum bands, colour presentations and investiture ceremonies. Affiliation of units with those in foreign armies it had fought alongside in the two World Wars, caste and ethnically specific regiments raised by the British, such as Sikh, Gurkha, Jat and Rajput, amongst a myriad others, were also likely to be discontinued for their colonial overhang. So were some Army insignias featuring British associations and symbols and institutes and roads in Indian cantonments named after English soldiers who had served the Empire in India and in other colonial outposts and wars.

Also likely to be scrapped in the Army is the long-established British tradition of appointing one-star officers and above as 'honorary colonels', or 'eminence grise' to their former battalions or regiments, in recognition of their services and experience. These appointments were akin to those of an 'agony aunt' for the unit and doing away with them, many veterans agreed, would only deprive the battalion of well-wishers who were often problem-solvers.

Furthermore, the Army's eight-odd regulation officers' uniform too is likely to undergo an *atmanirbharta* 'revision', with the more fancy and colourful ones, with English overtones, either being done away with or localised. These include the dashing blue patrols, the ceremonial winter uniform that comprises a black *bandgala* coat with gleaming silver-colour buttons, with each wearer's rank elaborately embroidered on the shoulders, and similarly coloured trousers.

But one facet that has surprisingly escaped the *atmanirbharta* tsunami and remained truly foreign in its entirety are the nicknames of the fighter, transport and helicopter squadrons of the three services. These monikers smacked of undiluted colonial influence, and consequently were ripe targets for either being rechristened with indigenous names, or subjected to direct but complex translations into Sanskrit, much like the IAF's catchy but dense squadron mottos.

Nicknames for IAF squadrons included Tigers and Wolfpack (Mirage-2000Hs), Winged Arrows (Su-30MKIs), First Supersonics (MiG-29Ms), Flaming Arrows (SEPECAT Jaguars), Flying Daggers (LCA Tejas), Skylords (Boeing Globemaster C-17 IIIs), Veiled Vipers (C-130J-30s) and Armoured Kestrels (Mi-17V5 helicopters), to name just a few.

The IN, for its part, had White Tigers and Black Panthers (MiG-29K/KUBs), the Condors (P-8Is), Winged Stallions (IL-38s), and Kestrels (Dhruv MkIII Advanced Light Helicopters), amongst others, while the Army Aviation Corps had Night Raiders, Blazing Falcons and Soaring Gideons (Dhruv ALHs) alongside Destroyers (Rudra-ALH Weapons Systems Integrated).

However, while decolonising India's cultural ethos and baggage is commendable, brutally purging historical ties can be counter-productive, especially for an institution like the military, for which tradition remains a sacrosanct lifeline. Such arbitrary moves also engender a backlash, especially amongst veterans, who are the backbone of the services and need a listening to. Besides, an august naval commanders' conference in the prevailing precarious security environment seems hardly the forum to debate sartorial matters with dressmaker's dummies in attendance; unless, of course, the endeavour was merely some good old English buffoonery.

Hope floats for augmentation of Navy's underwater assets

RAHUL BEDI
SENIOR JOURNALIST

THE prospects of France assisting India in developing and constructing six nuclear-powered general-purpose attack submarines or SSNs for the Indian Navy (IN) are believed to have brightened following a meeting between PM Narendra Modi and President Emmanuel Macron in New Delhi on the sidelines of the recently concluded G20 summit.

Discussions between the two leaders — who were meeting for the second time in two months following Modi's July visit to Paris as the chief guest at France's Bastille Day parade — were centred on the "joint development, manufacture and testing of advanced military platforms and technologies", said officials in Delhi. French assistance for the IN's continuingly deferred SSN programme has long been on the anvil, with hush-hush discussions over it at an advanced stage.

Earlier, the joint Horizon 2047 document agreed upon during Modi's Paris visit to mark 25 years of the bilateral strategic partnership between Delhi and Paris referred to both sides agreeing to "explore more ambitious projects to develop the Indian submarine fleet and its performance". Mazagon Dock Shipbuilders Limited in Mumbai had already licence-built six Kalvari (Scorpene)-class conventional diesel-electric 'killer-hunter' submarines under transfer of technology from France's Naval Group, and is poised to begin constructing three more to further boost collaboration in more advanced underwater platform production.

India had approached France for assistance with its then putative and floundering nuclear-powered missile submarine (SSBN) programme after Paris had robustly backed New Delhi's 1998 'Shakti' nuclear tests. And though this support, when India was globally isolated and under US-led sanctions, had led to the two inking the bilateral strategic partnership, France had declined to be involved in India's SSBN programme as, at the time, such assistance contravened prevailing global nuclear protocols. This negation had left India no choice but to turn to Russia as a default option, but senior industry officials conceded that the impending SSN project in the overall transformed security environment presented Paris and Delhi yet another collaborative opportunity to be exploited productively.

BOOST: India has added maritime competence to its land and air-based capacity for delivering strategic weapons. PTI

SSNs are employed for sundry missions such as anti-submarine and anti-surface warfare, while SSBNs perform specialised strategic nuclear deterrence missions. The latter's principal operational objectives are to safeguard a nuclear force's second-strike capability by offering a safe and undetected location from where missiles can be launched when required.

Earlier this year, France reportedly offered to jointly develop SSNs with India, under the aegis of Delhi's *Atmanirbharta* initiative designed to enhance self-sufficiency in materiel requirements, by transferring technology based on its Barracuda-class SSNs, the first of which — Suffren — was commissioned into the French Navy in mid-2022. Designed by the Naval Group, the 4.765-tonne and 99m-long Suffren was the first of the six SSNs scheduled to join France's Marine Nationale by 2030 at an estimated cost of over $2 billion each.

> As per the Horizon 2047 document, India and France agreed to 'explore more ambitious projects to develop the Indian submarine fleet and its performance'.

France was eager to clinch this prospective SSN deal with India to compensate for Australia summarily scrapping in late 2021 the multi-million-dollar deal with the Naval Group to supply the Royal Australian Navy 12 conventional submarines, for which the latter paid $585 million as compensation. Instead, Australia entered into a $268-368-billion deal with the US and the UK for eight SSNs between now and around 2050 under the AUKUS strategic pact.

The IN, for its part, is keen on partnering the Naval Group for its SSN needs as the technology it offered is more advanced compared to that of Russia, the only P5 country amenable to such sensitive cooperation with India. Industry officials and analysts are of the view that French involvement in India's SSN programme would, in all likelihood, be 'endorsed' by the US, whose strategic, defence and military technology ties with India are proliferating in a bid to challenge Chinese hegemony in the strategic Indian Ocean Region.

The IN's SSN project, worth an estimated Rs 60,000 crore, was approved by the Ministry of Defence in early 2015, with the first of the 6,000-tonne boats scheduled for completion by 2032-33. These platforms comprised part of the Navy's revised 30-year Project for the Series Construction of Submarines (till 2030) that envisaged the induction of 24 submarines, including six SSNs. These SSNs would supplement and operationally support the four or five 6,000-7,000-tonne Arihant-class SSBNs built with Russian knowhow and technical assistance, particularly with regard to miniaturising their 82.5MW pressurised light water reactors.

Meanwhile, India's SSBN programme, undertaken jointly by the Defence Research and Development Organisation, the Department of Atomic Energy and the IN, is proceeding apace at the secretive Ship Building Centre (SCB) in Visakhapatnam. The third such platform — simply designated as S4 — was launched in late 2021 after INS Arighat, the second SSBN, was undergoing further fitment. These SSBNs comprise a vital component of India's nuclear triad, aimed at sustaining its nuclear deterrence and no-first-use posture.

INS Arihant, the lead SSBN boat, was constructed at the SCB, drawing upon design elements of Soviet-era and Russian submarines from the Project 670A Skat-series ('Charlie I') and Project 667 ('Delta I') to the more recent Project 885 Yasen-class nuclear-powered cruise missile submarines (SSGNs). It was launched in July 2009; four years later, its on-board nuclear reactor attained criticality, enabling sea trials to begin in 2014 and weapon tests before its unannounced and low-key commissioning in August 2016. Two years later, Arihant completed its first deterrence patrol, confirming India's three-tier retaliatory nuclear deterrent capability by adding maritime strike competence to its existing land and air-based capacity for delivering strategic weapons.

The IN, however, which acknowledged the SSBN programme's existence only in December 2007, continues to claim that the Prime Minister's Office (PMO) had 'exclusive' management of the confidential programme. Funding for it was also confidential, supervised by the PMO via the National Security Adviser, it claimed. But having taken ownership of the SSN programme, it now behoves the Navy to imminently kickstart the project and stall the severe drawdown in its underwater assets.